

# The shadow\_hel Guard

Gating a cross-run transfer by spectral agreement

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## 1 Setup

Let  $s_1(p, r)$  be the round-1 (per-PSM) out-of-fold score for precursor  $p$  in run  $r$ . The cross-run “does this appear elsewhere?” feature is

$$\text{global\_max}(p, r) = \max_{r' \neq r} s_1(p, r').$$

It is a strong transfer signal but a *blind* one: a single confident hit in *any* other run yields a high value, even when  $p$  is truly absent from run  $r$ . That is the leakage it invites (yeast into human-only runs; a knocked-out gene into KO runs).

The missing signal is *spectral corroboration*. Write the top-8 smoothed fragment vector of the acceptor PSM as  $\mathbf{a} = (a_1, \dots, a_8)$  and of its *donor* (the run supplying the max,  $r^* = \arg \max_{r' \neq r} s_1(p, r')$ ) as  $\mathbf{d}$ . Normalize each to a probability vector,  $\tilde{a}_i = a_i / \sum_j a_j$ ,  $\tilde{d}_i = d_i / \sum_j d_j$ , and take the Hellinger distance

$$H(\mathbf{a}, \mathbf{d}) = \sqrt{\frac{1}{2} \sum_{i=1}^8 (\sqrt{\tilde{a}_i} - \sqrt{\tilde{d}_i})^2} \in [0, 1], \quad \text{agreement } A = 1 - H \in [0, 1].$$

A *true* transfer (same peptide, same fragmentation) has  $A \approx 1$ ; a *false* transfer (acceptor is noise) has  $A$  small.

## 2 Booster vs. guard

**Booster (current).** We hand the model `global_max` and `shadow_hel` =  $H$  as *separate additive features*. Gradient boosting is free to learn any monotone response, and it learns roughly “high `global_max` and high  $A \Rightarrow$  even more target-like.” Formally, the learned score  $F$  is (weakly) increasing in *both* arguments,

$$\frac{\partial F}{\partial \text{global\_max}} \geq 0, \quad \frac{\partial F}{\partial A} \geq 0,$$

so a large `global_max` can still promote a row at *low*  $A$ . Empirically this adds breadth but does not remove false transfers: agreement is used to *add* evidence, never to *veto* it.

**Guard (proposed).** We want disagreement to be able to *only subtract*. Two equivalent ways to impose this.

**Option 1 — gate the score.** Replace the raw feature by the *gated* score

$$\text{global\_max\_gated}(p, r) = \text{global\_max}(p, r) \cdot A = \text{global\_max}(p, r) (1 - H(\mathbf{a}, \mathbf{d}))$$

and feed *that* single feature to round-2 (no separate  $H$  term). Two properties make it a guard:

$$0 \leq \text{global\_max\_gated} \leq \text{global\_max}, \quad \text{global\_max\_gated} \xrightarrow{H \rightarrow 1} 0.$$

A borrowed score counts only to the extent the spectra agree. The residual false transfers are exactly the pairs with *large*  $\text{global\_max}$  (a real donor exists) but *large*  $H$  (the acceptor does not match it); the gate multiplies those toward 0, so they can no longer be rescued. A true transfer has  $H \approx 0$ , hence  $A \approx 1$ , and is passed through essentially untouched.

**Option 2 — monotonic constraint.** Keep  $H$  as a feature but constrain the trainer so the learned score is monotonically *non-increasing* in it,

$$\frac{\partial F}{\partial H} \leq 0 \quad \text{everywhere}$$

(LightGBM `monotone_constraints = -1` on that column). This is the same statement—disagreement can only lower the prediction—without hand-building the product; it lets the model choose *how* strongly  $H$  discounts, but never lets it help.

### 3 Why this is the right lever

**It targets the exact residual.** On the EWZ yeast/human benchmark, the selective-graft variant leaves +1,182 yeast identifications leaking into the human-only runs. Every such leak is a (high  $\text{global\_max}$ , low  $A$ ) pair by construction (the yeast donor is real; the human-only acceptor is noise). Option 1 maps precisely this set toward 0 while leaving the +13,156 genuine yeast gains (high  $A$ ) intact—so it should push the leak toward the +697 floor of plain `shadow+global_max` *without* sacrificing the +10.4% breadth.

**It composes with the shadow-decoy regularization.** The symmetric shadow-decoy makes each cross-run feature’s target/decoy marginal 1:1, forcing *joint* use with the base features. The guard is complementary: regularization stops the model from trusting the transfer score *alone*; the gate stops the transfer score from being *large at all* when the spectral evidence is absent. One acts on the estimator, the other on the feature.

**Interpretation.** The booster asks “is there evidence elsewhere?” The guard asks “is there evidence elsewhere *that matches what is actually in this run?*” Only the second question is false-transfer-safe, because a borrowed score can substitute for evidence, but a borrowed score *scaled by in-run spectral agreement* cannot exceed the in-run evidence.